

One-Page Research Paper Reproduction Summary

Paper: Deborah Pereg, "Back to basics: Fast denoising iterative algorithm," Signal Processing 221 (2024) 109482.

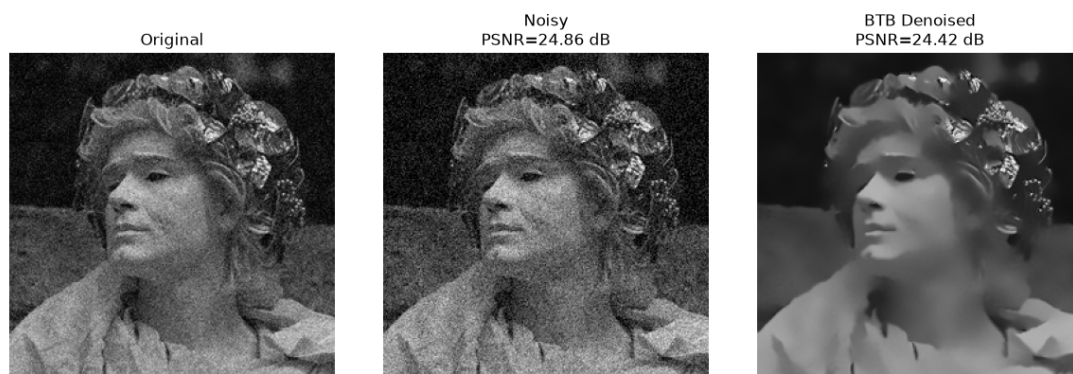
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Problem. The paper addresses image denoising when the degradation/noise level may be unknown. Its BTB idea repeatedly applies a denoiser and seeks a stable fixed-point-like solution.

Method reproduced. This OEL reproduces the natural-image AWGN workflow and the BTB update $x_{t+1} = (1-\mu)x_t + \mu f(x_t)$. A user-selected image is converted to grayscale, normalized, corrupted by AWGN, and processed iteratively. NLM is used as $f(\cdot)$ instead of the paper's pretrained TNRD, so this is a method-level reproduction.

Parameters. Main sigma = 15, sensitivity sigma = 10/15/25, T = 3, mu = 0.7, NLM patch size = 5, patch distance = 6, h = 0.8(sigma/255), image size = 256 x 256.

Evaluation. MSE, PSNR and SSIM quantify reconstruction quality. Additional graphs analyze sensitivity to noise level and iteration count.



Observed result. For the showcased image at sigma = 15, noisy-input PSNR is 24.86 dB and BTB + NLM PSNR is 24.42 dB. The output appears smoother, but the small PSNR reduction indicates that the fixed denoiser also removes reference detail.

Critical finding. The noise-level experiment shows improvement at stronger noise (sigma = 25) but over-smoothing at lower tested levels. The iteration experiment shows decreasing PSNR as iterations increase. Therefore, BTB performance depends strongly on the chosen base denoiser and parameters; more iterations are not automatically better.